Shoeb Bagwan

AI/ML Engineer

Mumbai, India | mohammedshoebbagwan@gmail.com | +91-8591780180

LinkedIn: shoeb-bagwan | GitHub: shoebbagwan

Summary

AI/ML Engineer with hands-on experience building Generative AI, NLP, and Retrieval-Augmented Generation (RAG) applications using Python, LangChain, and AWS. Skilled in developing end-to-end ML pipelines, from data preprocessing and model integration to deployment. AWS Academy Certified in Cloud Foundations and Data Engineering, with a track record of contributing to open-source projects and shipping deployed, working applications.

Technical Skills

Languages: Python, C++, SQL, HTML, CSS

AI/ML & NLP: Machine Learning Algorithms, Deep Learning, Artificial Neural Networks (ANN), Generative AI, NLP, RAG (Retrieval-Augmented Generation), Scikit-learn, LangChain, Google Gemini API, FAISS (Vector Search)

Frameworks & Libraries: FastAPI, React, Node.js, Pandas, Librosa, Streamlit, Leaflet.js

Databases & Cloud: SQLite, PostgreSQL, AWS (Cloud Foundations, Data Engineering), APIs

Tools: Git/GitHub, VS Code

Soft Skills: Problem Solving, Team Collaboration, Adaptability

Experience

Artificial Intelligence & Machine Learning Intern – Internship Studio Dec 2025 – Jan 2026

Remote

- Built an Artificial Neural Network (ANN) classification model in Python to classify Iris flower species from sepal and petal measurements, applying Principal Component Analysis (PCA) for dimensionality reduction, achieving high classification accuracy.
- Gained hands-on experience with the end-to-end ML workflow, including data preprocessing, feature scaling, model training, and evaluation.

Projects

RAG-Based PDF Chatbot *Python, LangChain, Google Gemini, FAISS, Streamlit*

- Built a Retrieval-Augmented Generation chatbot that answers questions strictly from the content of user-uploaded PDF documents, using semantic search over embedded document chunks to eliminate hallucinated answers.
- Deployed with Streamlit; handled real-world engineering constraints including API rate limits (batched processing) and embedding model migrations.

Vocal Metrics – Speech Emotion Recognition (SER) *Python, PyTorch, FastAPI, React, Librosa*

- Built a full-stack application that detects 8 distinct human emotions directly from raw audio waveforms, using acoustic signal properties rather than transcribed text.
- Extracted MFCCs (Mel-frequency cepstral coefficients) with Librosa for feature engineering and trained a PyTorch classification model, served via a FastAPI backend with a React frontend for real-time and pre-recorded audio input.

ChargePath – EV Charging Station Finder *React, Node.js, Leaflet.js*

- Developed a full-stack web application that helps electric vehicle users locate nearby charging stations with live availability status on an interactive map.
- Built the interactive mapping interface with Leaflet.js and a Node.js backend to serve real-time station data to the React frontend.

AttireAI – Personal AI Style Advisor *React, FastAPI, Scikit-learn, SQLite*

- Built a personal styling application that recommends outfits based on the user's body type, local weather, and occasion, using a Scikit-learn recommendation model served through a FastAPI backend.
- Designed the SQLite data layer and React frontend to deliver personalized recommendations end-to-end.

Open Source Contributions

- **django-email-learning** – Contributed docstrings to Django model classes (Merged PR #632).
- **easypaydirect-mcp** – Documented query/response fields in tool documentation (PR #14).

Education

Bachelor of Engineering in Computer Science (AI & ML) 2022 – 2026

Theem College of Engineering, Mumbai CGPA: 6.6/10 (60%)

Relevant Coursework: Data Structures, Algorithms, Database Management Systems (DBMS), Artificial Intelligence, Cloud Computing.

Certifications & Awards

- AWS Academy Cloud Foundations – March 2025
- AWS Academy Data Engineering – September 2025
- Green Skills and Artificial Intelligence – Edunet Foundation

Languages

English, Hindi